Introduction to Robotics and Intelligent Equipment

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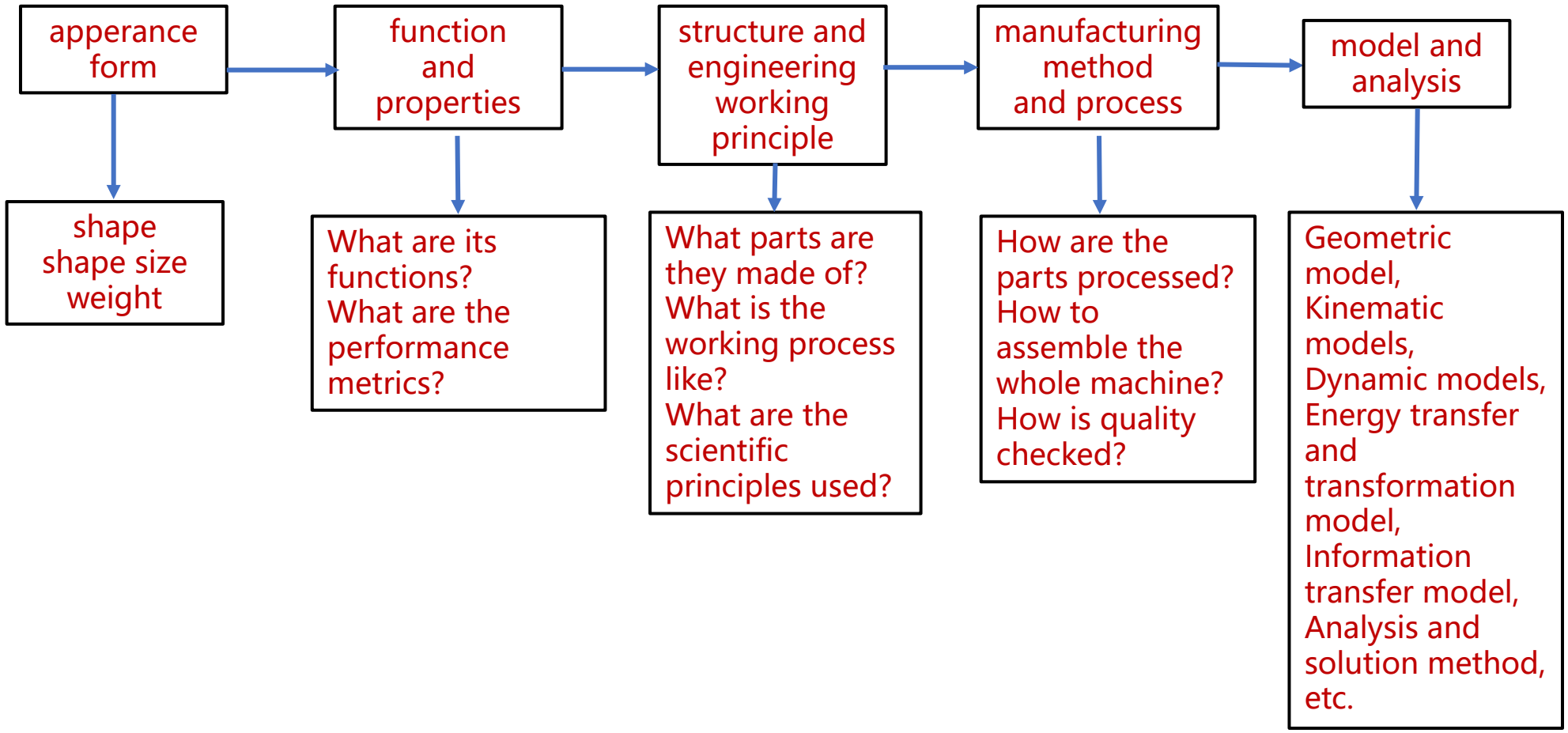
one

How to interpret electromechanical equipment



How to interpret electromechanical equipment

1. Understanding a mechanical device from shallow to deep



one How to interpret electromechanical equipment

2. Take the internal combustion engine as an example.

- **appearance shape**



- **Function** converts chemical energy of fuel into mechanical energy
- **performance**

✓ Dynamic performance indicators: how much power to send out, how much power/torque reserve.

✓ Speed range: minimum speed, maximum speed

✓ Economic performance index: fuel consumption per unit power unit time.

✓ Reliability and durability indicators: the longest operating time between major repairs or replacement parts and the ability to work without failure for a long time.

✓ Environmental performance indicators (NOx, HC, CO, particle, noise): harmful emissions per unit power unit time.

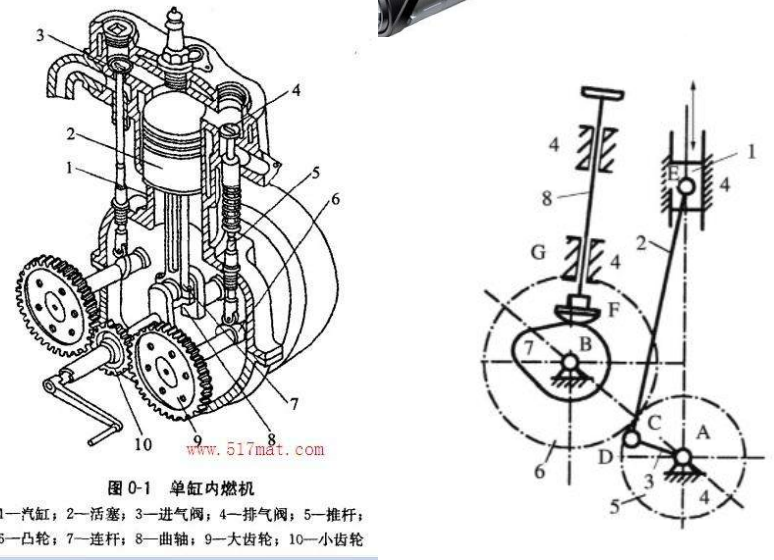


图 0-1 单缸内燃机

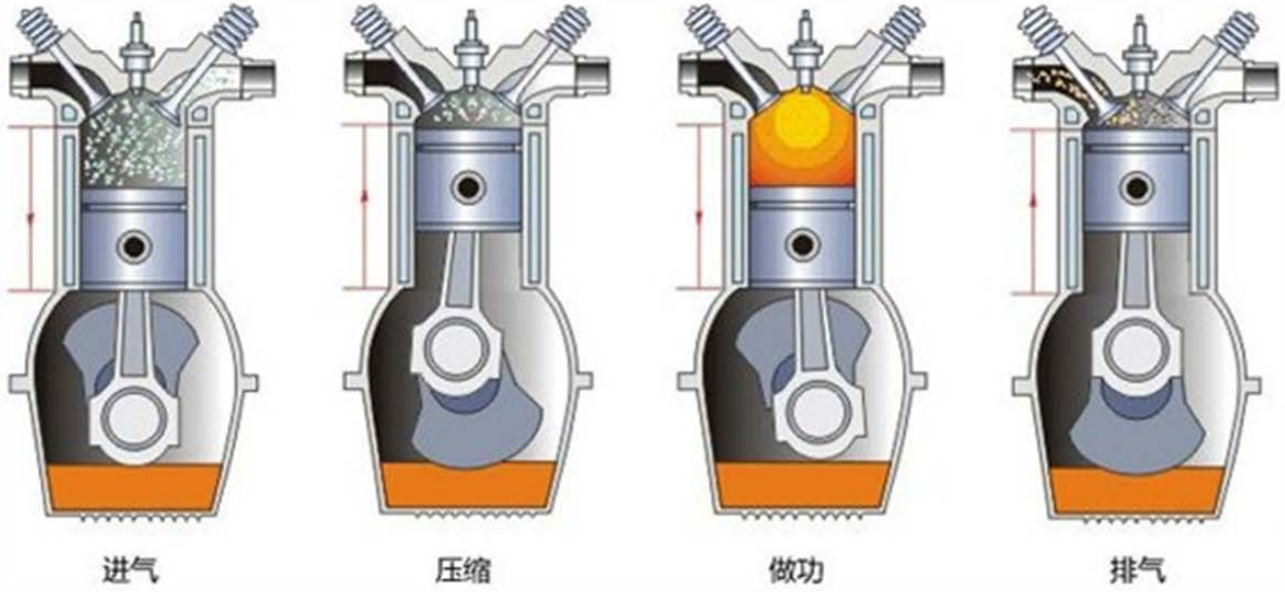
1—汽缸；2—活塞；3—进气阀；4—排气阀；5—推杆；6—凸轮；7—连杆；8—曲轴；9—大齿轮；10—小齿轮

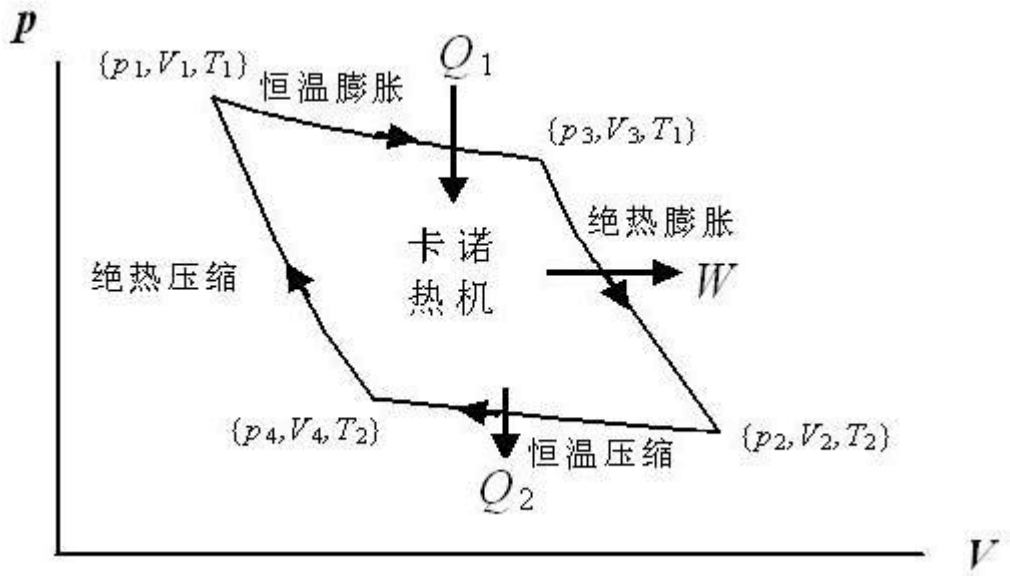
- **Basic Structure** Crank-connecting rod mechanism (cylinder, piston, connecting rod, etc.), valve train, fuel supply system, lubrication system, cooling system, ignition system, starting system

How to interpret electromechanical equipment

2. Take the internal combustion engine as an example.

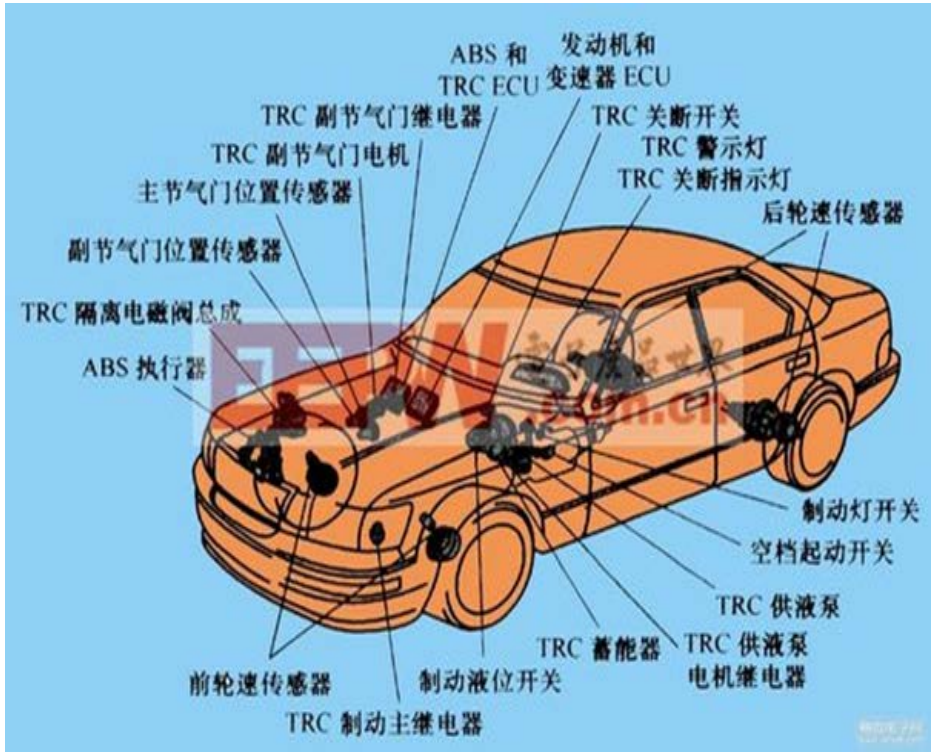
- **Working process** Four-stroke
- **Working principle** Carnot theorem, isovolumetric heating cycle (Otto cycle), isobaric heating cycle (Brayton cycle), mixed heating cycle (Diesel cycle).





two

One of the typical cases: car



two

One of the typical cases- car

1. development history



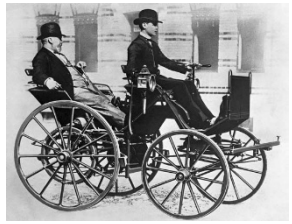
- **Steam engine** mounted on the scooter steam scooter, marking the use of machine-powered vehicles appear.

Late 18th century (birth)



19th century (growth)





- In 1885, German engineer Karl Benz developed a tricycle equipped with a 0.85 horsepower gasoline engine. The car is believed to be the world's first **motor** vehicle.
- In 1885, German engineer Gottlieb Daimler installed a gasoline engine on a bicycle and developed the first motorcycle.
- In 1886, Daimler installed his engine on a carriage made in the United States. This car is widely recognized as the world's first four-wheel gasoline engine.

20th century (mature)

- In 1908, Ford Motor Company produced the world's first ordinary car, the **Model T**.
- In 1913, Ford Motor Company developed the first assembly line in the automobile industry and began **mass production of** automobiles.
- They began to pay attention to the beauty and comfort of cars, and streamlined models such as the "Beetle" appeared.
- Japan's auto industry is growing.
- In 1889, the Frenchman Armand Bieru developed the **transmission** and **differential** device.
- In 1891, the French first adopted a front-engine rear-wheel drive and developed a friction plate **clutch**.
- In 1895, Scottish inventor John Dunlop invented the pneumatic **rubber tire**.
- The French Renault I in 1989 featured a **box transmission**,**universal joint** drive shaft and geared final drive.

21st century (new challenges)

- Zero pollution. Electric cars, solar cars, multi-fuel cars.
- Intelligent. a carrier for the comprehensive integration of many high and new technologies. Including navigation system, safe driving system, rescue system, unmanned driving, etc.
- New material. Lighter, faster and better.
- Beautiful exterior. According to the requirements of ergonomics, the design of body art modeling is carried out from the interior of the automobile.













two One of the typical cases- car

2. External form: shape, three-dimensional size, weight, type, etc.



3. Function: Overall function manned load, travel to destination. Specific functions: automatic emergency braking system, reverse imaging system, intelligent cruise control system, tire pressure monitoring system and automatic anti-lock braking system

4. Performance indicators:



1) Vehicle equipment quality (kg): The quality of the car fully equipped, including the quality of all devices such as lubricants, fuel, on-board tools, spare tires, etc.

2) Maximum total mass (kg): the total mass of the car when fully loaded.

3) Maximum loading mass (kg): The maximum loading mass of the car when driving on the road.

4. Maximum axle load mass (kg): The maximum total mass carried by a single axle of a car. Related to road traffic.



5) car length (mm): the distance between the two extreme points in the length direction of the car.

two One of the typical cases- car

6) vehicle width (mm): the distance between the two extreme points in the width direction of the car.

7) Vehicle height (mm): the distance between the highest point of the car and the ground.

8) wheelbase (mm): the distance from the center of the front axle to the center of the rear axle of the car.

9) Track length (mm): the distance between the center lines of the left and right tire treads of the same car.

10) Front suspension (mm): The distance from the front end of the car to the center of the front axle.

11) Rear suspension (mm): The distance from the rearmost end of the car to the center of the rear axle.

12) Minimum ground clearance (mm): the distance from the lowest point to the ground when the car is fully loaded.

13) Approach angle (°): The angle between the tangent of the front wheel of the car and the ground.

14) Departure angle (°): The angle between the tangent line of the rear end protruding point of the car to the rear wheel and the ground.

15) Turning radius (mm): When the car turns, the radius of the track circle of the center plane of the car's outer steering wheel on the vehicle support plane. The turning radius when the steering wheel is turned to the limit position is the minimum turning radius.

16) Maximum speed (km/h): The maximum speed that a car can reach when driving on a straight road.

17) Maximum climbing degree (%): The maximum climbing ability of the car when the car is fully loaded.

18) Average fuel consumption (L/100km): The average fuel consumption per 100 kilometers of cars driving on the road.

19) Number of wheels and number of driving wheels (n×m): The number of wheels is the measurement basis, n represents the total number of wheels of the car, and m represents the number of driving wheels.

two One of the typical cases- car

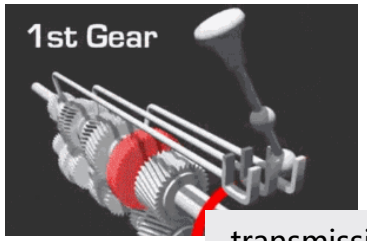
5. Structure and working principle

- **Support part:** chassis + frame
- **Power part:** engine, or electromagnetic + inverter + motor
- **Transmission part:** electric gearbox + differential + clutch, universal joint, etc.
- **Control part:** steering wheel, throttle, brake, gear lever and other units

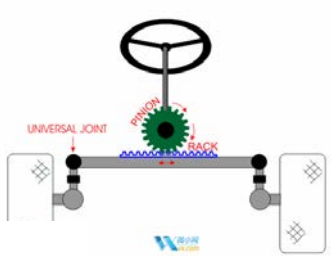




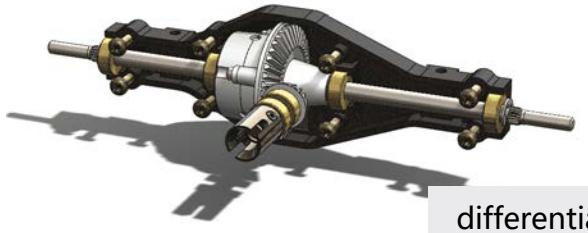




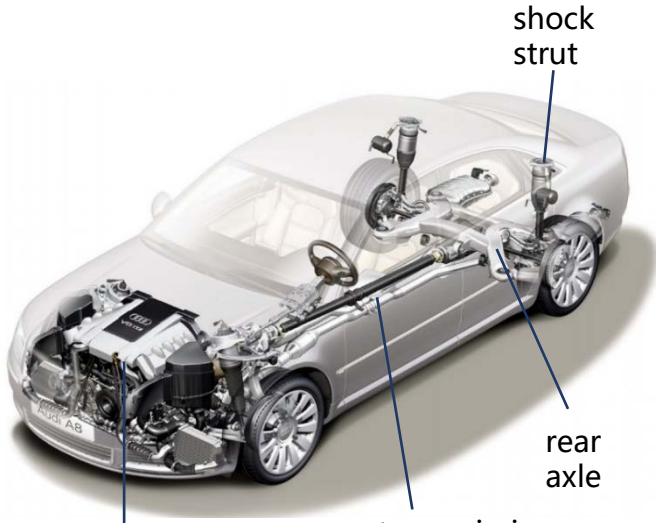
transmission

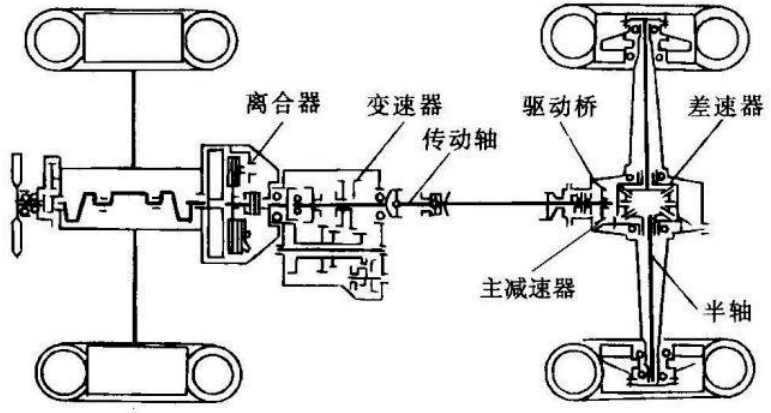


knuckle mechanic



differential

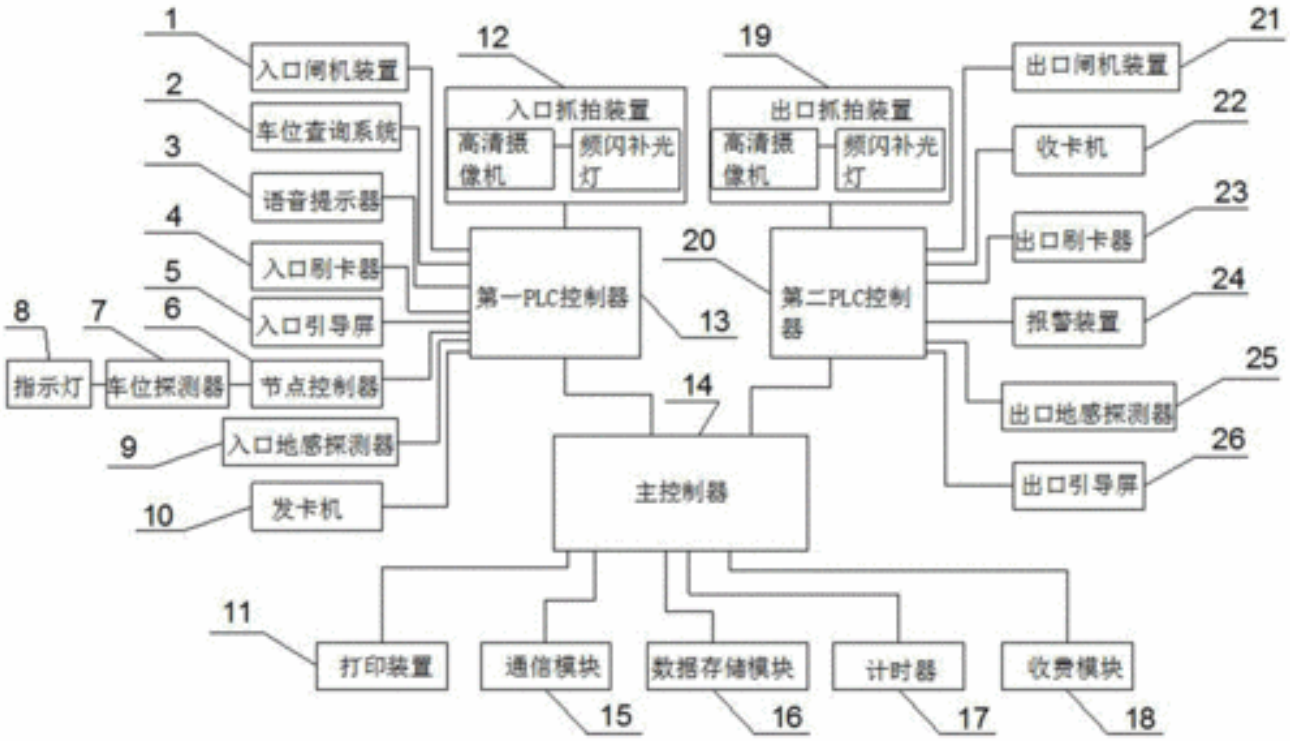




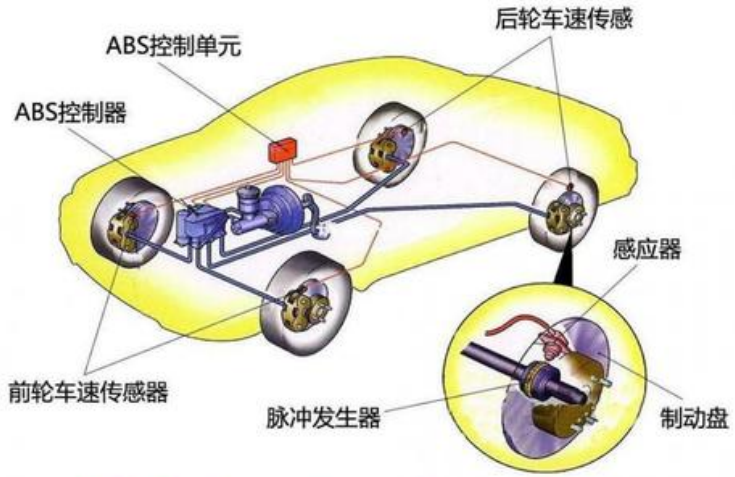
two One of the typical cases- car

5. Structure and working principle

- **Control part:** computer +IO+ motor driver + sensing system
- **Executive part:** wheel + brake parts + air conditioner + sound, etc.







ABS系统的组成及布置

two One of the typical cases- car

5. Structure and working principle

■ **How it works:**

The car is injected by ECU, and the high-voltage package will generate high-voltage electricity to supply spark plug. After gasoline and air are mixed, the electric explosion of spark plug will push spark plug, spark plug will ignite and explode, generate power, and drive wheel rotation through transmission system.

Engine: The engine is the power plant of the car, its role is to make fuel combustion to generate power, and then through the chassis transmission system to drive the wheels of the car to travel.

Chassis: chassis role is to support, install the automobile engine and its various parts, assembly, forming the overall shape of the car, and accept the power of the engine, so that the car produces movement, to ensure normal driving.

Body: The body is mounted on the frame of the chassis for the driver, passenger or cargo. Car, passenger car body is generally the overall structure, truck body is generally composed of cab and cargo box two parts

How the car works
https://b23.tv/ulOQvv0

One of the typical cases- car

6. The impact of cars on social development

Key Points

- safety design
- reliability design
- environmentally compatible design (Low consumption, low emission)
- Human-Computer Interaction design

POSITIVE

- Caused a revolution in production management
- forming a network of roads.
- accelerated urbanization.
- Exploited the automobile service trade
- A whole new set of things came into being.

NEGATIVE

- Excessive resource consumption
- serious environmental pollution
- Frequent traffic accidents
- complex social problems



three

Typical Case II: CNC Milling Machine

three

Typical Case II- CNC Milling Machine

1. External form: shape, three-dimensional size, weight, type, etc.

2. Function: Cut off the excess material on the surface (usually non-rotating surface) of the workpiece by means of tool rotation and workpiece movement.

3. Performance indicators:

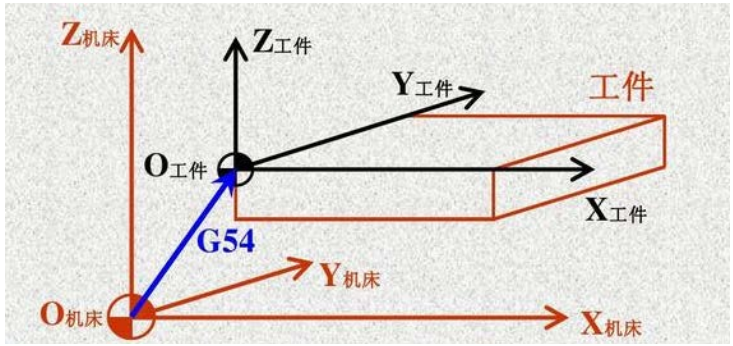
1) Motion performance: spindle speed, travel range of each motion axis, maximum and minimum feed speed, movement stroke size, etc.

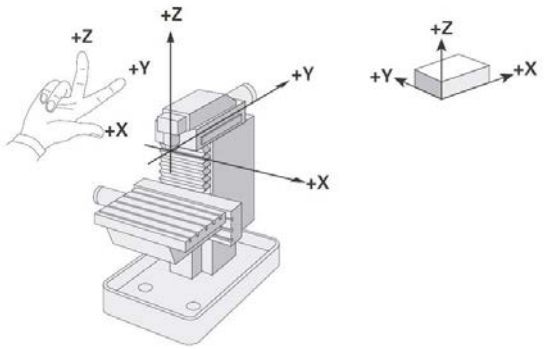
2) Accuracy index: positioning accuracy, repeated positioning accuracy, displacement resolution, etc.

3) Coordinate axis control index: controllable axis number, linkage axis number.

4) Cutting ability index:material removal rate per minute.







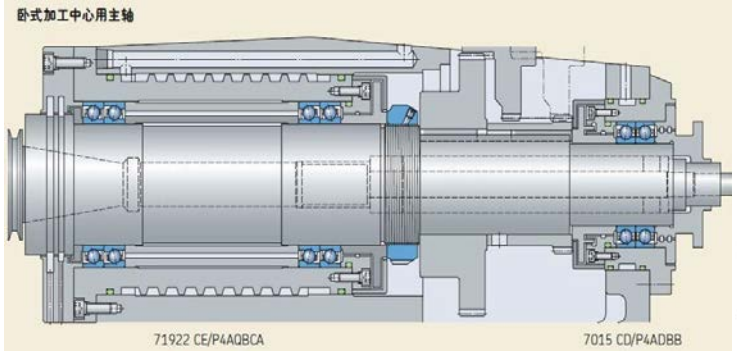


three

Typical Case II- CNC Milling Machine

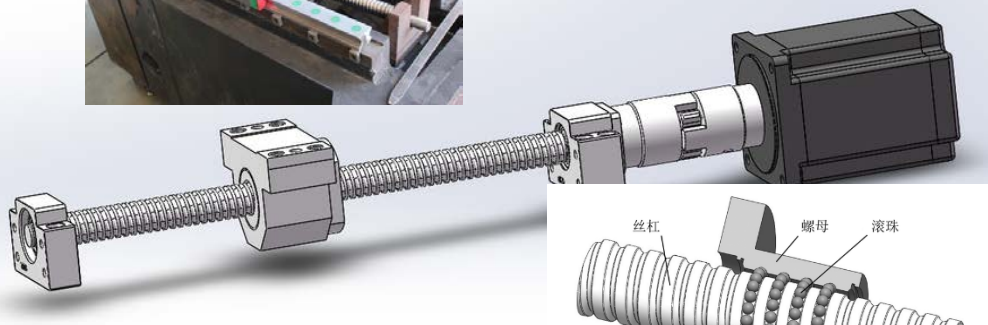
4. Structure and working principle

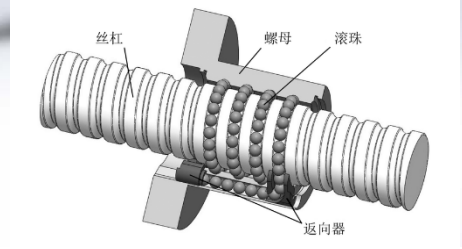
- **Support part:** bed body
- **Power part:** electric motor
- **Transmission part:** lead screw, etc.
- **Manipulation part:** operation panel, etc.



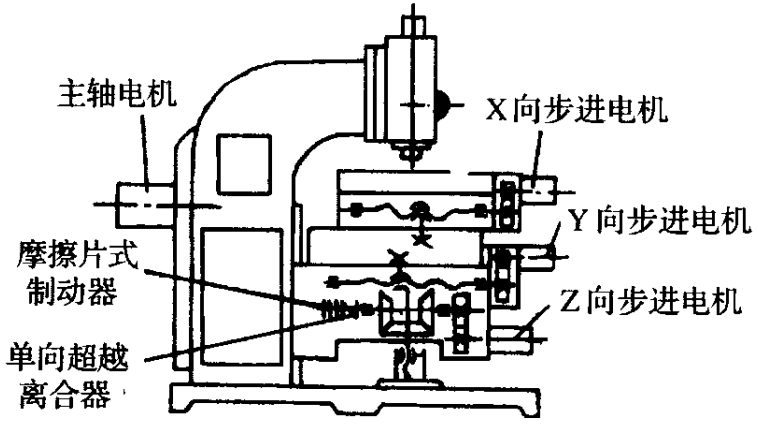












Typical Case II- CNC Milling Machine

4. Structure and working principle

- **Control part:** computer +IO+ motor driver + sensing system
- **Execution part:** worktable + spindle + tool of each motion axis

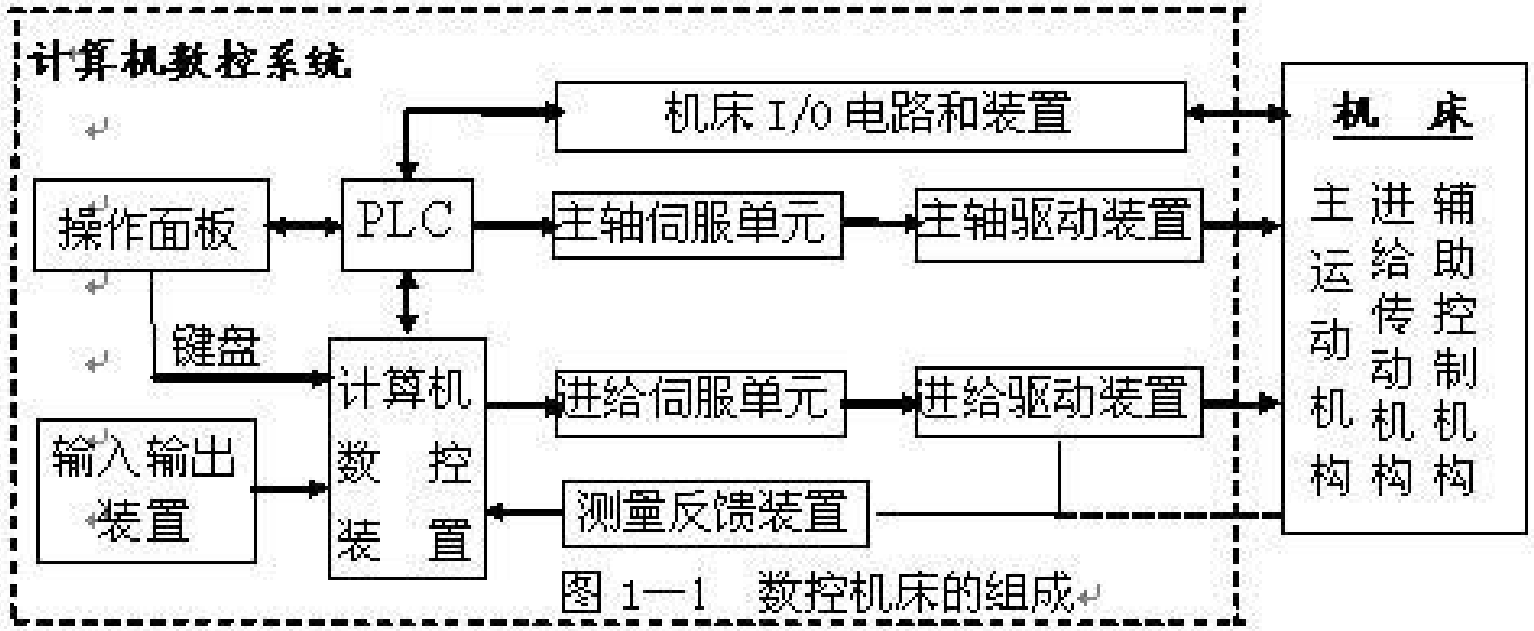
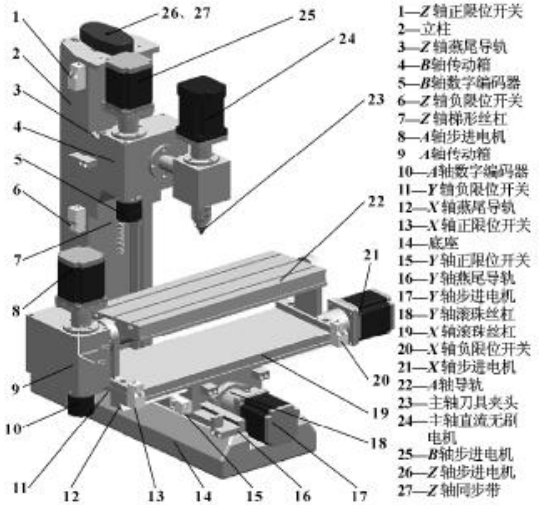
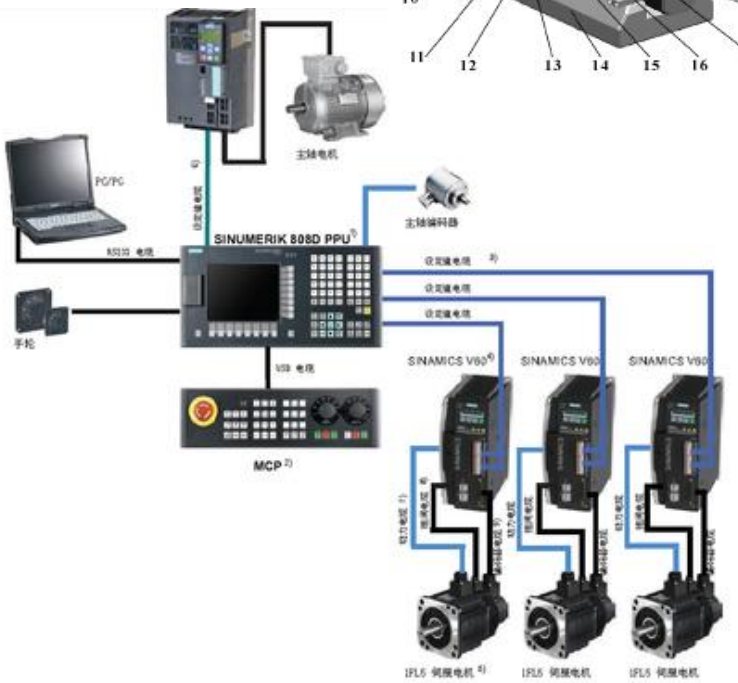


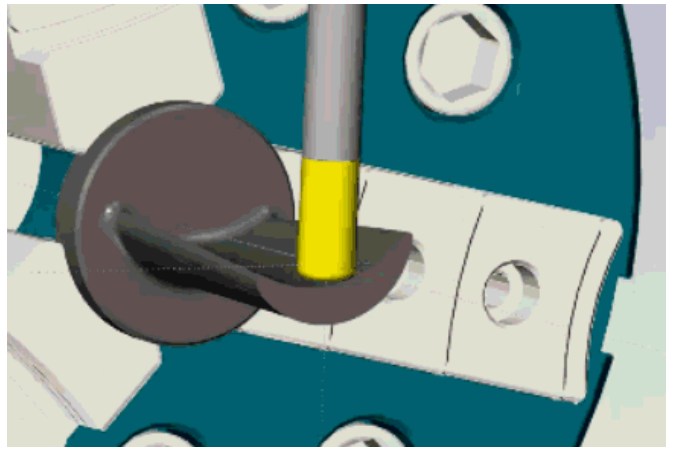
图 1—1 数控机床的组成

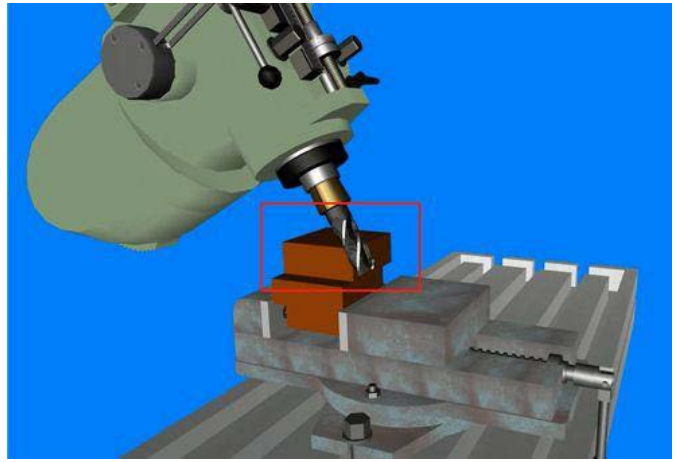




three Typical Case II-CNC Milling Machine

4. Structure and working principle

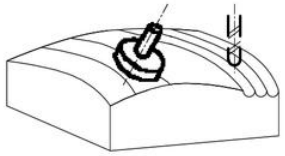


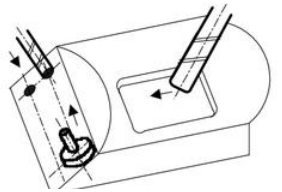


- **Working principle:** according to the part shape, size, precision and surface roughness and other technical requirements to develop processing technology, select processing parameters. Through manual programming or automatic programming with CAM software, the programmed machining program is input to the controller. After processing the machining program, the controller transmits instructions to the servo device. The servo device sends control signals to the servo motor. The spindle motor rotates the tool, and the servo motors in X, Y and Z directions control the relative movement of the tool and the workpiece according to a certain trajectory, so as to realize the cutting of the workpiece.
- **Two core and critical mechanisms:**

Continuous cutting mechanism: tool rotation provides cutting speed and power, table movement provides feed speed and power.

The mechanism of trajectory control: Solve the motor control parameters according to the trajectory and relative motion speed of the tool point.

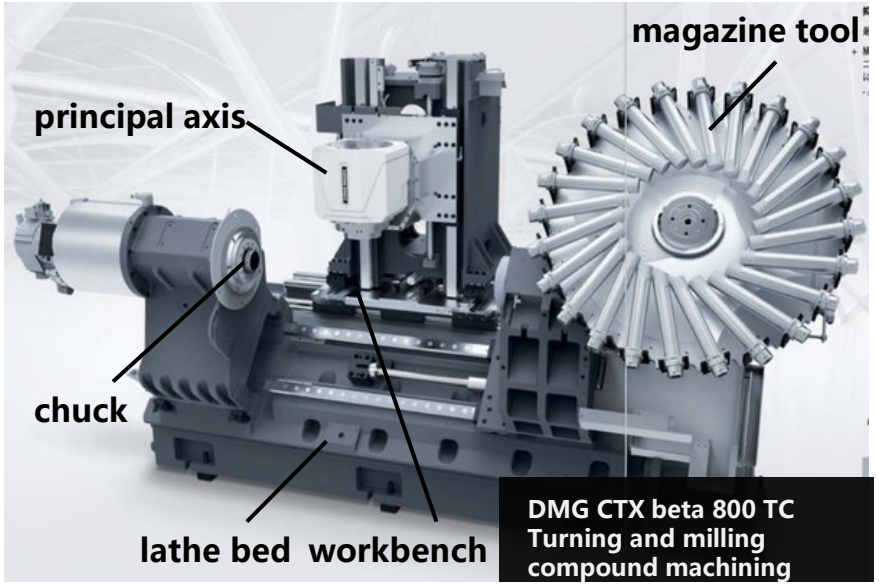


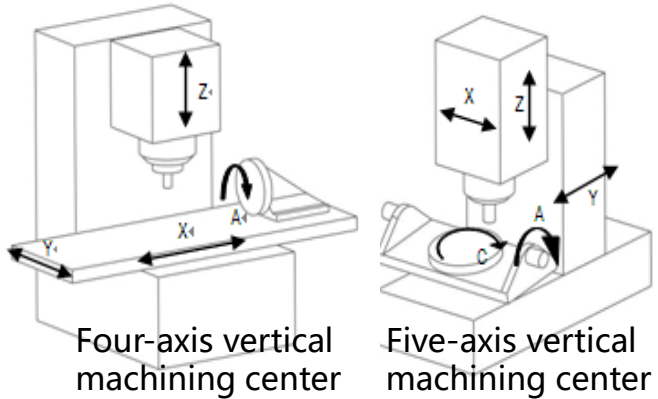


Typical Case II-CNC Milling Machine

5 Introduction to machining center

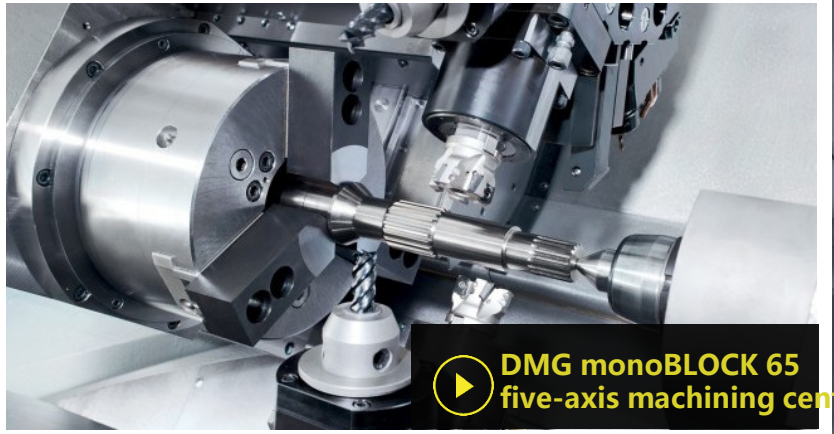
- By adding a tool magazine to the machine tool, it is possible to complete as many automatic machining tasks as possible after one clamping. A machining center is defined as follows: A machining center contains at least three CNC feed axes and includes an automatic tool changer and a CNC machine tool with a tool storage device.
- Classified according to the number of feed shafts, it can be divided into three-axis machining centers, four-axis machining centers and five-axis machining centers.
- Classified according to the position of the spindle, it can be divided into horizontal machining centers and vertical machining centers.

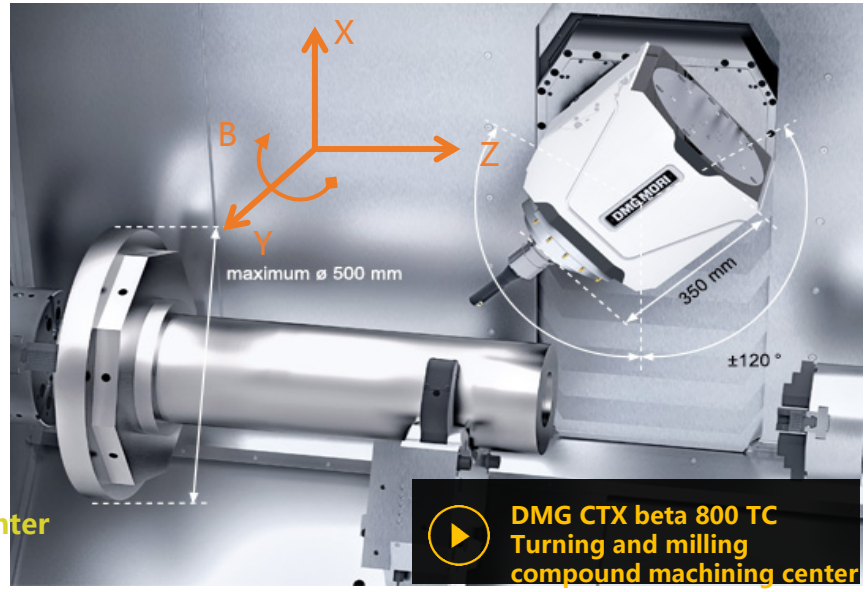




Four-axis vertical machining center

Five-axis vertical machining center





Typical Case II-CNC Milling Machine

6. Function and Significance of Developing NC Machine Tools and Machining Centers

Importance and function of machine tool and machining center

Machine tool processing range expanded

It can process parts with complex profiles that are difficult to complete with ordinary machine tools or that cannot be processed at all. In particular, the development of compound machining machines and five-axis machining centers supports the machining of turbine blades and propeller blades in a single fixture.

Improve production efficiency and reduce labor costs

CNC machine tool is a highly automated, high-efficiency machine tool, with automatic transmission, automatic tool change, automatic exchange of workpieces, productivity has been greatly improved, generally 3-4 times that of ordinary machine tools, or even higher. For some complex parts, the productivity can be increased dozens of times in the processing of large and error-prone parts, such as circuit board processing.

Fast response to production changes

CNC machine tool processing parts replacement, production preparation is relatively easy, production preparation cycle is also short. The machining center adopts process concentration mode to realize multi-station, multi-surface and multi-tool machining in one part positioning and clamping. Make the speed of responding to production changes fast, improve the economic benefits of enterprises.

High precision machining

CNC machine tool itself has good rigidity, high precision and good precision retention, which is conducive to the long-term stability of CNC machine tool processing quality. Accuracy correction and compensation can be performed using software. The machining accuracy of ordinary CNC machine tools has been improved from ±10μm to ±5μm and ±2μm, and the accuracy of high-precision machine tools can reach ±0.2μm. In ultra-precision machining, the machining accuracy reaches the nanometer level.

Good for modern management

CNC machine tool processing parts, can accurately calculate the product production hours, and effectively simplify the inspection, fixture and semi-finished product management work; the use of CNC information standard code input, so easy to connect with the computer, constitute a computer-controlled and managed production system, to achieve the modernization of manufacturing and production management.

Easy maintenance and repair of equipment

High-end CNC systems generally have a self-diagnostic alarm function, and display the approximate scope and nature of the fault.

What is CNC Machining and How Does it Work?
https://www.bilibili.com/video/BV1Ls411g7sw/?share_source=copy_web



four

Typical Case 3: Industrial Robots

four Typical Case 3- Industrial Robot

1. External form: shape, three-dimensional size, weight, type, etc.

2. Function: replace people to complete a certain job in the industrial production process, such as: loading and unloading in the machining process, workpiece handling, welding, cleaning, painting, etc.

3. Performance indicators:

1) Degrees of Freedom: The number of axes of independent motion that the robot has.

2) Load bearing capacity: the maximum load that can be carried. Load-bearing capacity is an important index to evaluate the working ability and application range of robots.

3) Working range: The size of the effective working area of the industrial robot depends on the configuration, joint number and joint structure of the robot.

4) Repeated positioning accuracy: The accuracy that industrial robots can achieve when performing the same task is an important parameter to measure the stability and accuracy of robot movements.





Typical Case 3-Industrial Robot

3. Performance indicators:

5) Speed: Speed refers to the speed at which the robot moves when performing tasks.

6) Acceleration: The time required for the robot to reach its maximum speed from a stationary state is an important parameter for evaluating the flexibility and responsiveness of the robot.

7) Control accuracy: The control accuracy that the robot can achieve when performing tasks, including motion tracking, position control and force control.

8) Dynamic performance: ability to accelerate and decelerate, and ability to respond to external disturbances.

9.) Electricity demand: A certain amount of electricity is required for work. Power demand usually includes parameters such as rated voltage, rated power and power requirements.

10) Safety performance: Safety needs to be considered during operation. Safety features include robot protection mechanisms, safety sensors and emergency stop devices.

Typical Case 3-Industrial Robot

3. Performance indicators:

11) Control system: The control system of industrial robot is the core component of robot to realize precise movement and perform tasks, including robot controller, programming mode and communication interface.

12) Stability: The anti-disturbance ability and motion stability of industrial robots when performing tasks are important indicators for evaluating robot motion accuracy and work quality.

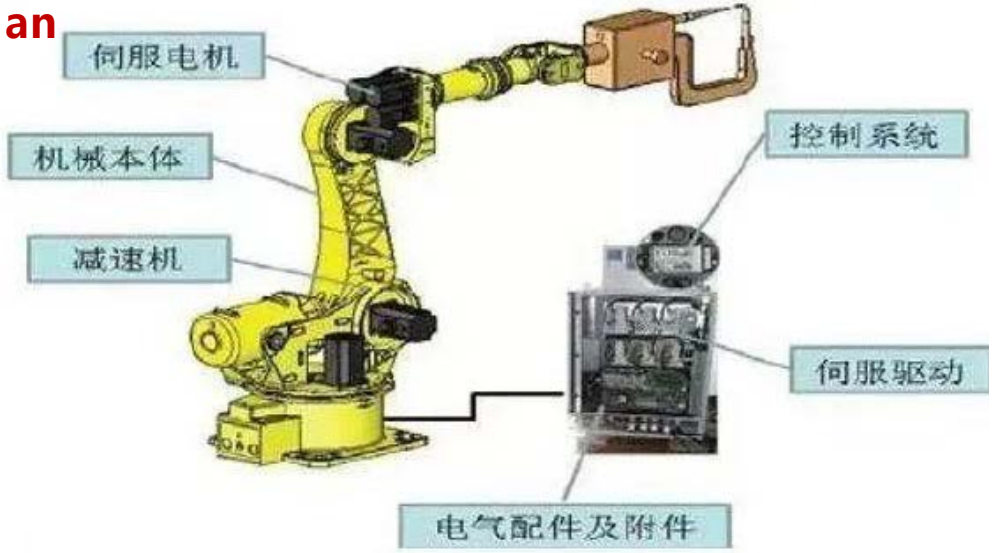
13) Life and reliability: The performance and reliability of industrial robots during long-term use. The evaluation of lifetime and reliability includes maintenance cycle, failure rate and maintainability of robot.

14) Applicable industries and applications: The applicable industries and application scope of industrial robots are also important factors in evaluating their technical parameters.

four Typical Case 3-Industrial Robot

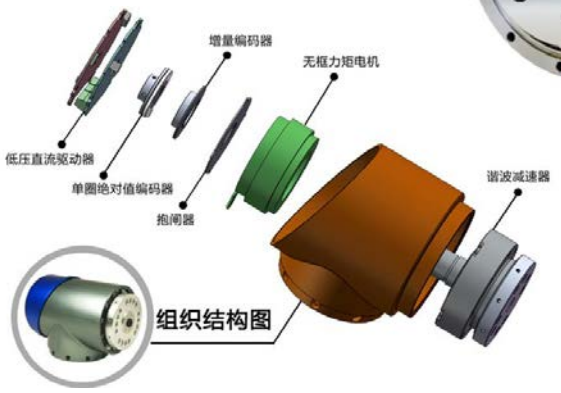
4. Structure and working principle (taking light articulated robot as an example)

- **Support parts:** fixed base, mobile body, robot arm, etc.
- **Power part:** motor, etc.
- **Transmission part:** reducer or hydraulic transmission device, etc.
- **Manipulation part:** remote control, operation panel, et









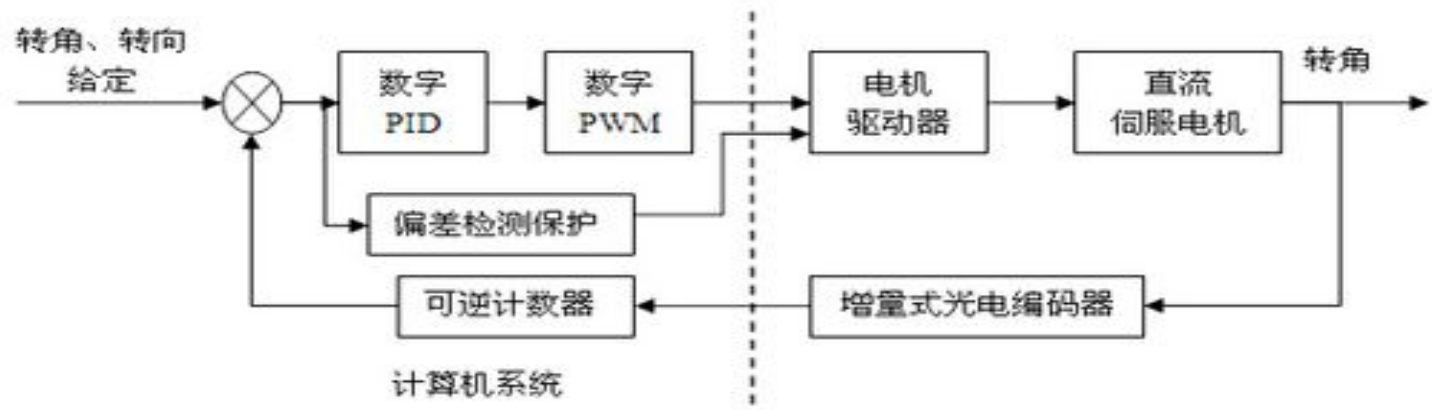


Typical Case 3-Industrial Robot

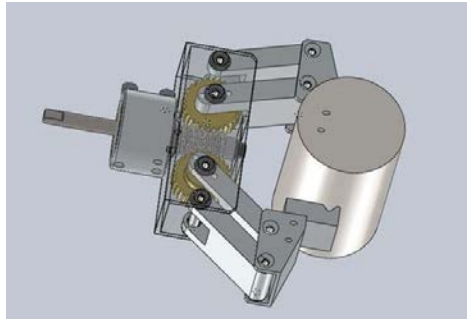
4. Structure and working principle

- **Control part:** computer +IO+ motor driver + sensing system
- **Execution part:** robot end effector, such as: robot arm, welding gun, spray gun, etc.













4. Structure and working principle

- **How it works:**

The working principle of robots, including perception, decision-making and execution. robot senses relevant information and parameters of surrounding environment and system through equipped sensors, completes processing of information, computer system makes corresponding control decisions according to information processing results and preset task requirements, operation planning, relevant rules and algorithms, etc., and then decomposes decision results into instructions for controlling each moving part and operation part according to these decisions, robot movement and composition of operation parts, and sends them to corresponding drivers and actuators. Therefore, the robot can be controlled to complete the corresponding operation actions and tasks, such as moving, grasping, rotating, etc.

The working principle of robot can also be understood from the working principle of robot control system and robot mechanical execution system. The control system is the robot's brain, responsible for the entire process of perception, decision-making and execution. The mechanical execution system is responsible for the execution of instructions and the completion of work tasks.

Typical Case 3- Industrial Robot

5. Role and significance of developing industrial robots

- **Positive effects:**

1. Improve production efficiency: The high automation and precision of robots can greatly improve production efficiency, reduce production costs and improve corporate profitability.

2. Replace people to complete the work that people do not want to do or can not do: robots can replace some repetitive, monotonous or dangerous work, reducing the dependence on human resources.

3. Improve product quality: Robots can perform very fine work without human error, thus improving product quality.

4. Operator safety: Robots can replace some dangerous jobs, reduce the possibility of operator injury, and ensure operator safety.

four Typical Case 3-Industrial Robot

5. Role and significance of developing industrial robots

- **Negative effects:**

1. Job losses: As the use of robots in the industrial field continues to expand, it will lead to the loss of some labor positions.

2. Safety issues: For example, failure or loss of control of a robot may result in casualties or property damage.

3. Ethical questions: should robots replace doctors in decision-making? Should robots be held responsible for medical malpractice?

4. Data privacy issues: robot applications require a large amount of data support, but the data collected by robots also has the risk of privacy leakage. For example, household data collected by smart home robots may be accessed by unauthorized third parties.

four Typical Case 4-Humanoid Robot

Boston Dynamics
https://www.bilibili.com/video/BV1Mv411N7zq/?share_source=copy_web

Tesla Optimus
https://b23.tv/MDdJ2Ko

Thank you!

